SE Lab 04 DinoGame (code changes the tasks implemented)

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Task 1: Fix the high-score bug

Bug:

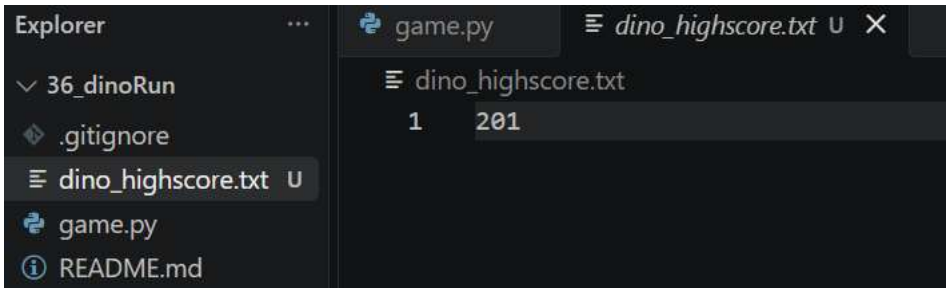
```
if self.high_score > current:
    self.high_score = current
    save_high_score(self.high_score)
return
```

Fix:

```
if current > self.high_score:
    Self.high_score = current
    save_high_score(self.high_score)
return
```

Explanation:

If the current run score is greater than the high score then it becomes the high score and should be displayed rather than 0. Now the high score can be seen updated in the dino_highscore.txt doc also



Task 2: Implement random colour when the dino is in air

Bug:

```
def dino_tint(on_ground):
    pass
```

Fix:

```
def dino_tint(on_ground):
    if not on_ground:
        return(random.randint(0,255),random.randint(0,255),random.randint(0,255))
    return None
```

Explanation:

The old code gives the same colour for the dino in air as well as ground but this updated function gives it a random colour while dino in air i.e, not on ground

Task 3: Add a jump “+1” after every obstacle crossed

Bug:

- def on_obstacle_passed(obstacle,score):
  pass
- self.obstacles=[o for o in self.obstacles if not o.is_off_screen()]
- for obs in self.obstacles:
  obs.draw(screen)

Fix:

- Add popups list
- def on_obstacle_passed(obstacle,score):
  popups.append([obstacle.rect.centerx,obstacle.rect.top-10,30])
- self.obstacles=[o for o in self.obstacles if not o.is_off_screen()]
  for popup in popups:
  popup[1]-=1
  popup[2]-=1
  popups[:]=[p for p in popups if p[2]>0]
- for obs in self.obstacles:
  obs.draw(screen)
  for x,y,life in popups:
  surf=self.font.render("+1",True,(50,50,50))
  screen.blit(surf,(x,y))

Explanation:

Creates a new popups list which stores all the current popup information whenever an obstacle is crossed, we do popup[1]-=1 and popup[2]-=1 because once jumped we need the obstacle to move up so y and time changes.

Task 4: Double jump addition

Bug:

```
def max_jumps():
    pass
```

Fix:

```
def max_jumps():
    return 2
```

Explanation:

Allows only one jump for the old code but return 2 sets the max jump to 2 jumps helping to use jump while in air.